Class-Incremental Learning Network for Small Objects Enhancing of Semantic Segmentation in Aerial Imagery

Junxi Li, *Graduate Student Member, IEEE*, Xian Sun, *Senior Member, IEEE*, Wenhui Diao, Peijin Wang, Yingchao Feng, *Graduate Student Member, IEEE*, Xiaonan Lu, *Graduate Student Member, IEEE*, and Guangluan Xu

***Abstract*— Due to the differences in the feature distribution between classes, when the model learns in a continuous data stream, it will encounter catastrophic forgetting. The incremental learning methods have shown great potential to solve this problem. However, most existing methods based on task-incremental learning are difficult to adapt to characteristics of remote sensing scenes with few differences in appearance but large differences in features, which is not conducive to artificially distinguish task-identity document (ID). Thus, we propose a class-incremental learning (CIL) network for small objects enhancing semantic segmentation in aerial imagery. Specifically, considering the superior accuracy of the binary classifier, we propose a twin-auxiliary (TA) model that adds an auxiliary binary classification task. Then, for expansion and contraction at the edge and small object confusion problems, we introduce a diversity distillation loss, using the results of binary-classifier to constrain the multiclass segmentation results and strengthen the attention to the locations of the segmentation results that have changed. Finally, we design a conflict reduction mechanism for multihead classifier to achieve single-head prediction for CIL. Experiments demonstrate that our method has good performance on the Vaihingen and Potsdam datasets by the International Society for Photogrammetry and Remote Sensing (ISPRS), outperforming state-of-the-art (SOTA) incremental learning methods. The code will be available soon.**

***Index Terms*— Catastrophic forgetting, edge contraction, incremental learning, knowledge distillation, small-target confusion.**

I. Introduction

SEMANTIC segmentation is one of the most essential applications in the field of remote sensing image interpretation. It has great significance to urban monitoring, land cover detection, and other fields. However, most models need to get all the data at once to achieve excellent performance, which is not in line with the rapid update iteration of remote sensing data. When the model is trained in the form of a data stream, it will suffer the catastrophic forgetting [2]–[5] problem that refers to a significant decrease in the performance of previous data due to changes in model parameters. The reason for the catastrophic forgetting problem is that different data categories have different distributions, which is shown in Fig. 1. On the one hand, different settings of sensor parameters lead to domain differences in data under different imaging conditions. On the other hand, there are differences in pixel distribution of different types of object samples under the same imaging conditions. In this case, incremental learning [6]–[8] is considered to be the most promising way in order to ease above problems.

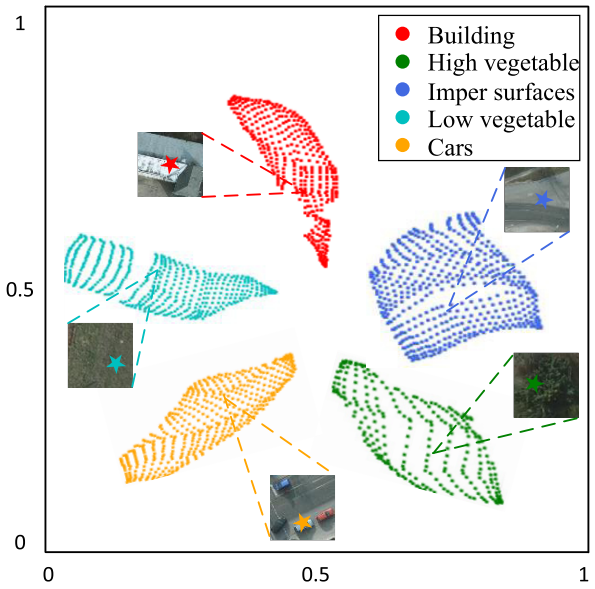


Fig. 1. t-distributed stochastic neighbor embedding (t-SNE) [1] visualization results of Vaihingen dataset. Each color represents one category. The features dimensions are reduced from 512 to 2, where the abscissa and ordinate, respectively, represent the feature 1 and feature 2 after dimensional reduction. The t-SNE is a method for dimensionality reduction and visualization of high-dimensional data. Imper surfaces refers to impervious surfaces.

Manuscript received August 10, 2021; revised October 11, 2021; accepted October 27, 2021. Date of publication October 29, 2021; date of current version February 14, 2022. This work was supported by the National Natural Science Foundation of China under Grant 61725105. *(Corresponding author: Xian Sun.)*

Junxi Li, Xian Sun, Yingchao Feng, and Xiaonan Lu are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, also with the University of Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100190, China (e-mail: lijunxi191@mails.ucas.ac.cn; sunxian@aircas.ac.cn; fengyingchao17@mails.ucas.edu.cn; luxiaonan19@mails.ucas.ac.cn).

Wenhui Diao, Peijin Wang, and Guangluan Xu are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China (e-mail: diaowh@aircas.ac.cn; wangpj@aircas.ac.cn; gluanxu@mail.ie.ac.cn).

Digital Object Identifier 10.1109/TGRS.2021.3124303

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

Most existing incremental learning methods rely on task-incremental learning [2], [9]–[14] that refers to the incremental learning method which requires entering the task-identity document (ID) into the model in inference time. According to the difference of tasks, task-ID aims at distinguishing different datasets or categories. Moreover, the accuracy of the task-incremental learning system is quite dependent on the task-ID. But it should be noted that it is hard for people to identify the task-ID to which it belongs accurately during validation. Especially for remote sensing images with few differences in appearance but large differences in features. For example, it is difficult for the naked eye to distinguish between trees and low vegetation because they are green and irregular in shape. The task-ID preset manually may be incorrect, so the task-incremental learning system is not suitable for remote sensing. Therefore, avoiding task-ID is an important condition for incremental learning, which is called class-incremental learning (CIL).

In order to realize CIL, various methods have been proposed in the past few years [7], [15]–[18]. The main challenges to achieving CIL are the following two aspects: one is to distinguish between all classes from all previously learned tasks without using the task-ID. The other is to realize the learning of new tasks while retaining the performance of the old tasks. Most of the previous works [17], [18] retain partial samples of the old tasks, and classify all the learned categories by repeatedly training feature extractors and classifiers. However, our method achieves CIL without retaining the old data. In this case, our method can be applied to scenarios where data is more private. For the second challenge, the knowledge distillation is considered to be an effective way to retain the knowledge of the old tasks. Current methods [7], [19] apply knowledge distillation [20], [21] to avoid catastrophic forgetting but ignore the characteristics of the remote sensing images. Inspired by the incremental learning for semantic segmentation (ILSS) method [7], we are committed to exploring the main causes for catastrophic forgetting of remote sensing images. We found that applying the traditional knowledge distillation method to achieve incremental learning of sensing images still has the following problems.

1) *Expansion and Contraction at the Edge:* Limited by the resolution of remote sensing images, the edge features of remote sensing images are relatively unclear. When the new data arrive, ILSS directly uses the output logits as dark knowledge for knowledge distillation. There is no constraint on the corresponding edge position. Therefore, it is hard for the network to remember that the pixels of the old data have been classified (the pixels of old tasks are the background). In the incremental learning process, it is easy to cause edge expansion and contraction, as shown in Fig. 2(a) and (b).
2) *Confusion at Small-Scale Objects:* Compared with natural scene datasets, there are more small-scale objects in remote sensing datasets, such as cars. If the classes of new data contain small-scale objects, the classes with a large number of pixels learned in the old task and subsequent tasks will be treated as the background. This will cause an unbalanced number of foreground and background pixels. The ILSS method calculates the binary cross-entropy loss for each pixel and outputs the binary image corresponding to each category, which causes a more serious front-background imbalance problem. Therefore, in the process of CIL, it is easy to misclassify small-scale objects into other categories, which in turn affects the classification accuracy, as shown in Fig. 2(c) and (d).
3) *Semantic Distribution Shift of the Background Class:* in the process of achieving incremental learning in semantic segmentation, the pixels of the old tasks are set as the background class when the new data arrives. Similarly, the pixels that were previously set as the background class may now appear like the foreground class. The change in the distribution of pixels contained in the background class of the same classifier is known as the semantic distribution shift of the background class. Meanwhile, the output probabilities of pixels corresponding to the background class change according to the increase of the categories, which will result in a more serious catastrophic forgetting problem. The schematic of semantic distribution shift of background class is shown in Fig. 3.

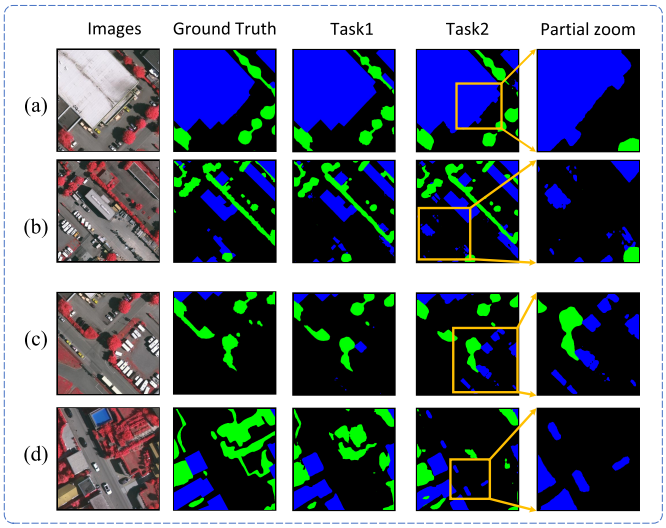


Fig. 2. Cause of catastrophic forgetting. The first column represents input images. The second column represents the ground truth (GT) of the corresponding image of task1. The third column indicates the segmentation prediction result of the image after task1 is completed. The fourth column is the predicting result of the pixels belonging to task1 in the image after the training of task2 is completed. The last column represents the partial zoom of the fourth column. (a) and (b) Catastrophic forgetting occurred in expansion and contraction at the edge. (c) and (d) Catastrophic forgetting at the location of small-scale objects.

Regarding the limitations of existing knowledge distillation methods for remote sensing characteristics, we consider optimizing the incremental learning methods based on knowledge distillation. We introduce auxiliary tasks to constrain the position of edges and small objects. By focusing on the positions that are difficult to learn, the model can ensure that the positions that are hard to learn are paid more attention to. Besides, because the multihead model can avoid semantic

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

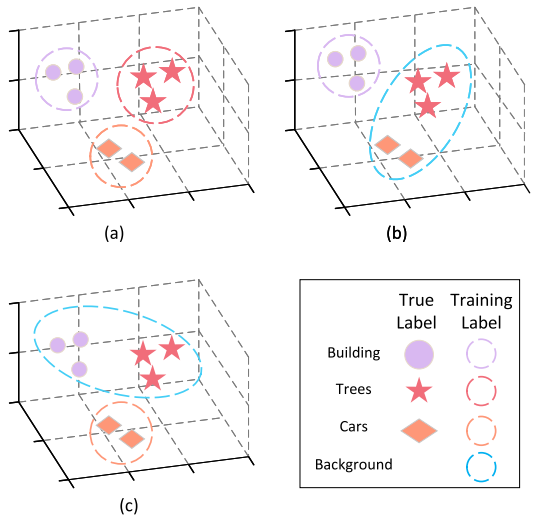


Fig. 3. Illustration of semantic distribution shift of background category. (a) Static learning (SL). (b) Task1. (c) Task2. In the comments: the first column represents the category, the second column is the actual category in the image, and the third column indicates the label category. In SL, the actual label and the training label are completely corresponding. In task1, the labels of the input image are only buildings and background. In task2, the labels of the input image are the car and the background. In each incremental step, the pixels where the label is the background change all the time.

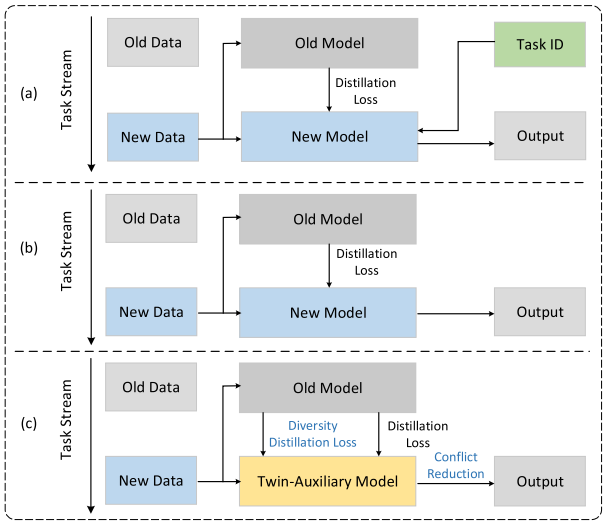


Fig. 4. Schematic of task-incremental learning pipeline, CIL pipeline, and ours. (a) Task-incremental learning pipeline. (b) CIL pipeline. (c) Our proposed CIL pipeline.

shifts of the background class, our model uses multihead classifiers. Each classifier predicts the foreground and the background, where the predicted foreground should be the pixels classified by the current task, and the predicted background should be the pixels classified by the previous task. However, due to the lack of task ID, each pixel location may be classified into foreground categories at the same time. In order to integrate the foreground content of different output classifiers, we propose a conflict reduction strategy, using single-head prediction to test the multihead model.

Inspired by the above thoughts, we propose the CIL framework, which consists of a twin-auxiliary (TA) model, a diversity distillation loss, and a conflict reduction mechanism. The difference among task-incremental learning, CIL, and ours is shown in Fig. 4. Based on the characteristics of remote sensing datasets, we design the diversity loss function, which solves the confusion problem for small-scale objects and edges in the incremental learning process, and reduces the forgetting of small objects and the edge of objects. Experiments demonstrate that the effectiveness of our approach on the Vaihingen and Potsdam datasets provided by the International Society for Photogrammetry and Remote Sensing (ISPRS) [22]. The method achieves state-of-the-art (SOTA) results on the Vaihingen and Potsdam datasets.

To summarize, the contributions are as follows.

1) By analyzing the main reasons for the catastrophic forgetting problem in remote sensing images, we propose a CIL framework consisting of a TA model, a diversity distillation loss, and conflict reduction, which can realize CIL of successively increasing data without utilizing the task-ID as prior information.
2) Aiming at the edge expansion and contraction problem and the small objects' confusion problem of the old tasks for CIL in aerial imagery, a diversity distillation loss function is proposed to constrain the edges and small objects that are easy to forget in the old tasks.
3) To achieve single-head evaluation, we propose a conflict reduction mechanism to solve the problem of the same pixel is divided into multiple categories when predicting label fusion.

II. Related Works

In this section, we specifically review the relevant methods of incremental learning [23] that also referred as continual learning, sequential learning [24] or lifelong learning [25]. Early incremental learning focused on studying task-incremental learning tasks and then gradually exploring the CIL tasks.

A. Task-Incremental Learning

The problem of catastrophic forgetting has been discovered for many years. Already in the 1980s, McCloskey and Cohen [3] showed that while human subjects suffered from gradual forgetting of previously learned tasks, algorithms trained with backpropagation suffered from catastrophic forgetting. Most early incremental learning methods of the classification problem requires multihead testing based on different datasets, and the task-ID is an indispensable option. De Lange [16] grouped previous works in three categories: replay-based, regularization-based, and parameter isolation-based. In regularization-based methods, one of the earliest works is learning without forgetting (LwF) [10], in which the loss of knowledge distillation is used to preserve information of the old classes and this method needs to save the old data. Elastic weight consolidation (EWC) applies the Bayesian framework to neural networks, which allows determining the posterior distribution of parameters by introducing uncertainty on the network parameters, not just the point estimation in the parameter space [2]. Rotate-EWC (R-EWC) approximately diagonalizes the Fisher information matrix of the network

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

parameters and takes the form of factor rotation of the parameter space [26]. When combined with EWC (assumed to be a diagonal Fisher information matrix), it can be significant to improve the performance of lifelong learning for continuous tasks. Incremental moment matching (IMM) [27] performs IMM of the posterior distribution of the neural network trained on the first task and the second task, respectively, and assists in various transfer learning methods to solve the catastrophic forgetting problem. In replay-based methods, gradient episodic memory (GEM) [11] focus on minimizing negative backward transfer (catastrophic forgetting) by the efficient use of episodic memory. As an improved version of GEM, averaged GEM (A-GEM) [28] has not only the same performance as GEM but also reduces the complexity and the number of parameters. In parameter isolation-based methods, PackNet [29] utilizes pruning to free parameters for training new tasks.

B. Class-Incremental Learning

The CIL system discards the tedious task-ID prior knowledge, so it is more challenging than the task-incremental learning system. The representative work is incremental classifier and representation learning (iCaRL) [17], which realizes incremental learning by selecting and storing the samples closest to the feature mean of each class, and repeatedly training the feature extractor and the classifier. Large-scale incremental learning (LSIL) [18] uses the same sample storage method as iCaRL, which solves the problem of unbalanced data distribution in each class when the sample size is large. End-to-end incremental learning (EEIL) [30] introduces a balanced fine-tuning (FT) process after model training, and uses herding selection method to retain old data. The learning without memorizing (LWM) method proposed by Dhar *et al.* [31] uses attention distillation to punish the change of the attention map, thereby helping to retain the old information. Our article does not compare with the method of preserving old data or old data-related content, because we believe that this is not in line with the ideal situation of incremental learning.

C. Incremental Learning Beyond Image Classification

Early research on incremental learning focused on image classification. With the development of incremental learning, researchers apply incremental learning to object detection and semantic segmentation. In the field of object detection, the core of the solution proposed by Shmelkov *et al.* [32] is a loss function to balance the classification loss and the distillation loss. The distillation loss minimizes the difference between the soft label distributions of the teacher network and the student network. Liu *et al.* [33] explored object detectors that continuously learn in different scenarios in different fields and categories, and proposed attentive feature distillation (AFD), which uses bottom-top and top-bottom attention to reduce forgetting.

Inspired by LwF, incremental learning techniques (ILTs) [19] apply common distillation loss to retain the knowledge of the old task without maintaining the data of the old task. ILSS [7] achieved incremental learning on semantic segmentation datasets of remote sensing for the first time. ILSS applies binary cross-entropy loss function, reducing catastrophic forgetting by distilling the output logits. However, it is not ideal to introduce the distillation loss used in incremental learning of classification into incremental learning of semantic segmentation. It does not consider the relationship between pixels, that is, the data of the new task not only contain the pixels of the current task, but also the pixels of the old task which serve as the background.

Subsequently, Cermelli *et al.* [34] discovered and proposed the semantic distribution shift problem of background class. That problem means that the pixels of the current task that are identified as background may be classified into a certain category in future tasks as the number of tasks increases. Cermelli *et al.* [34] eased the semantic distribution shift of background class by modifying the cross-entropy loss and distillation loss, which changes the distribution of output probabilities by adding up the predicted probabilities of all the old classes and the background probabilities. Our method avoids the problem of semantic transfer of background classes by adopting a multiclassifier model. Moreover, we propose new ideas and implementation methods for the problem of catastrophic forgetting.

D. Semantic Segmentation

Since the deep neural network was proposed, the accuracy of semantic segmentation has made a leap [35]–[47]. In 2015, Long *et al.* [48] proposed fully convolutional network (FCN), which uses 1×1 convolution instead of the fully connected layer to achieve pixel-level classification of input images and solve the end-to-end semantic level segmentation issues. Since then, researchers have made a lot of improvements to the FCN. Dilated convolution [46], [49] and pyramid pooling module (PPM) [50] contributed to enlarge the receptive field. The U-shaped network (U-Net) series methods [51], [52] using the encoder–decoder structure also achieved good results. In particular, TreeUNet [53] is the first adaptive convolutional neural network (CNN) to be applied to semantic labeling tasks of remote sensing images. The authors use an adaptive network to increase the classification rate at the pixel level. Segmentation transformer (SETR) [43] deployed a pure transformer to encode an image as a sequence of patches and model global context information at each layer of the transformer. But these methods are all carried out when all the data are obtained. When the data arrive sequentially in the form of a data stream, the model cannot adapt to this new type of data input form, which will result in a significant reduction in model performance. In our work, we proposed an incremental learning network, which has gradually increasing segmentation capabilities for successively increasing data.

III. Methodology

This section describes our method in detail. First, we define the problem and notations in Section III-A. Then, we introduce our class-incremental model named the TA model. After, the enhancing edge and small-objects incremental module (EESIM) in Section III-C aims at improving the performance of distillation in incremental learning at small targets

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

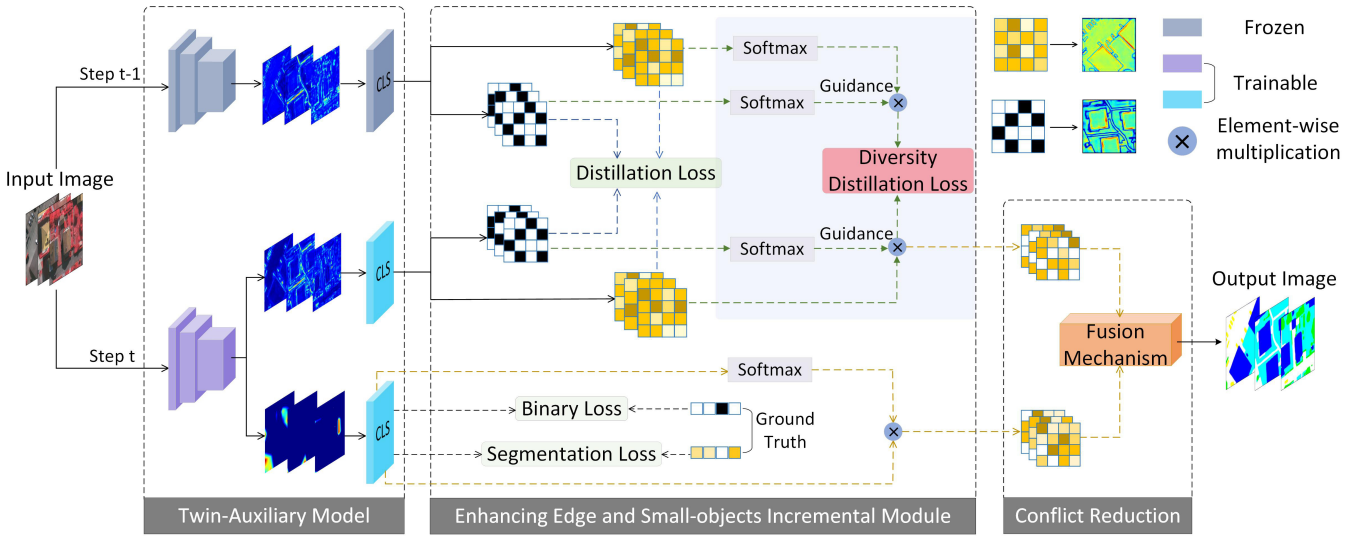


Fig. 5. Pipeline of our framework. It consists of a TA model, an enhancing edge and small-objects incremental module (EESIM), and a conflict reduction. During the training at tth step, the new TA model is trainable, while the old model is frozen. The training images of new task are fed into the old and the new models to obtain the corresponding mapping outputs, the binary loss and the segmentation loss is applied to learn new task, the distillation loss, and the diversity distillation loss are applied to preserve the knowledge of old tasks. During testing, we design a fusion mechanism to fuse the predicted labels of all tasks to get the final output result. We do not store any data for old tasks. The input images sent to the new and the old models are exactly the same.

and edge locations. Then, we describe the conflict reduction mechanism in Section III-D. Last but not least, we give the total loss function used to train the model in Section III-E.

A. Problem Definition and Notation

As data continue to emerge, the ability of the model to continuously learn from the increasing data is of more practical value. The distribution differences of different types of target samples under the same imaging conditions cause catastrophic forgetting. The incremental learning method has a good learning ability for successively increasing data and can effectively alleviate the catastrophic problem. The ideal incremental learning is carried out using only the knowledge of the current task, and our method is also designed based on this.

Let us denote $(0, 1, \ldots, t, \ldots, T)$ as different steps during incremental learning. The number of classes included in each task is not fixed. During the tth steps, the datasets are denoted as

$$D^t_{\text{train}} = \left(X^t_{\text{train}}, Y^t_{\text{train}}\right) \tag{1}$$

where $X^t_{\text{train}} = \{x_i\}_{i=1}^{n_t}$ contains the input images of size $H \times W \times 3$ for current task and $Y^t_{\text{train}} = \{y_j\}_{i=1}^{n_t}$ is the corresponding GT.

Specifically, Y^t_{train} only contains N^t kinds of different classes and class c^t_{bg} represents background class. But it should be noticed that the model should predict all the classes seen during the test. In our experimental settings, we do not compare with the replay-based method because keeping old data is not fit for ideal incremental learning settings. Meanwhile, our experiment does not use any GT from the old task data, thus $Y^t_{\text{train}} \cap Y^{t-1}_{\text{train}} = \varnothing$.

When training incremental learning model f_{θ_t} at tth step, semantic segmentation aims at predicting a label y_u for each pixel u of image x, where $y_u \in [0, 1, \ldots, N^t]$. Let us denote the training set as $\tau \subseteq (X \times Y)$, the input space X is mapped to class probability vector through the model f_θ. The output is denoted as $y^* = \text{argmax}_{c \in y_u} f_\theta(x)[u, c]$, where $f_\theta(x)[u, c]$ is the probability for class c of pixel u in image x. During the test, the model at tth step must be able to discriminate between all the classes that have been seen so far. As in standard CIL, we assume that the labels of all categories in the experiment are disjoint, except for the background class c^t_{bg}.

B. TA Model

At present, the incremental learning methods [19], [34] of semantic segmentation adopt the general segmentation models [51], [54]–[56]. However, it is difficult to alleviate the catastrophic forgetting problem of small-scale targets and object edges in remote sensing images by only adopting such a model structure. Therefore, we propose an incremental learning framework, as shown in Fig. 5. Our framework contains three components, namely TA model, enhancing edge and small-objects incremental learning module, and conflict reduction.

During the training at tth step, the new TA model is trainable, while the old model is frozen. The training images of the new task are fed into the old and the new models to obtain the corresponding mapping outputs, the binary loss and the segmentation loss is applied to learn new task, and the distillation loss and the diversity distillation loss are applied to preserve the knowledge of old tasks. During testing, we design a fusion mechanism to fuse the predicted labels of all tasks to get the final output result.

The TA model for the second task is shown in Fig. 6. In order to adapt to the new classes added when new tasks arrive, the existing methods are divided into two types of structure changes: one is to add an output classifier, and the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

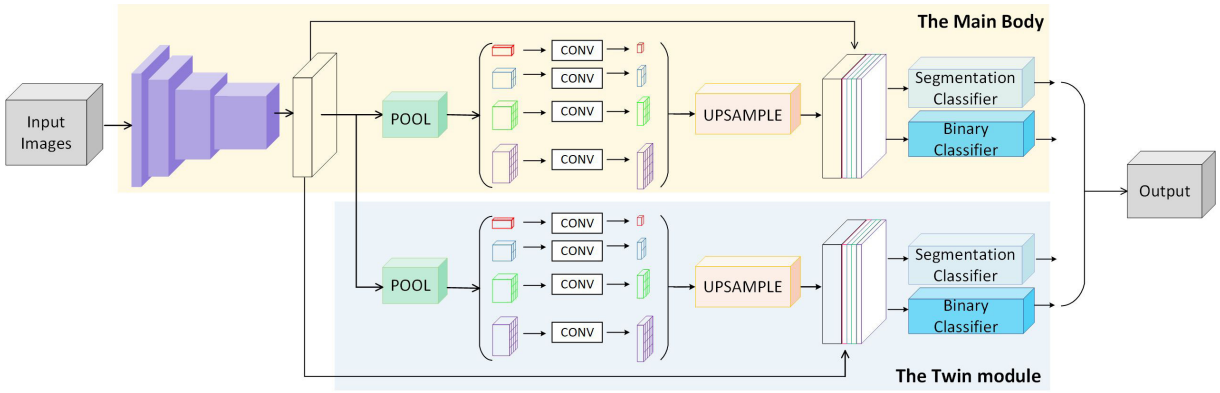


Fig. 6. Schematic of TA model for the second task. The TA model contains the main body, the twin module. The position with yellow background indicates the main body. The position with blue background indicates the twin module. Each twin module contains two classifiers, including binary classifier and segmentation classifier. The purpose of this is to add an auxiliary task to explore the pixel location where the predicted label is wrong.

other is to add nodes based on the original output layer. For semantic segmentation, adding nodes to the original output layer will lead to semantic distribution shift of background class. Therefore, we adopted the first method to design our model that is a CIL model.

In particular, the TA model consists of the main body and the twin module. The main body of the TA model is pyramid scene parsing network (PSPNet) [50] that is composed of multiple components. We choose Resnet50 [57] as the feature extractor because of its powerful feature extraction ability.

The twin module contains a PPM and two classifiers. The incremental learning process will degenerate into a common semantic segmentation problem when there is only one task. When the second task arrives, in order to retain both high-level semantic information and low-level semantic information, the network retains the PPM and initializes the number of output classes of the classifier according to the number of classes added by the new task.

Considering that the two-category segmentation task is simpler than the multicategory segmentation task, the two-category segmentation task has higher classification accuracy. Therefore, we propose to add another classifier named binary classifier in order to achieve two-category segmentation. The binary classifier contributes to exploring the pixel location where the predicted label is wrong. The position of the binary classifier is in parallel with the segmentation classifier, which is used to divide the input image into foreground and background. When the prediction results of the binary classifier and the segmentation classifier conflict, the model's attention is focused on this part of the area.

C. Enhancing Edge and Small-Objects Incremental Module

The EESIM uses multiple loss functions while retaining the knowledge of the old task and learning the knowledge of the new task. On the one hand, common distillation loss in the incremental learning process is difficult to avoid the catastrophic forgetting problem caused by small target confusion and edge shrinkage. So, we propose a diversity distillation loss to improve this phenomenon, which is shown in Fig. 7.

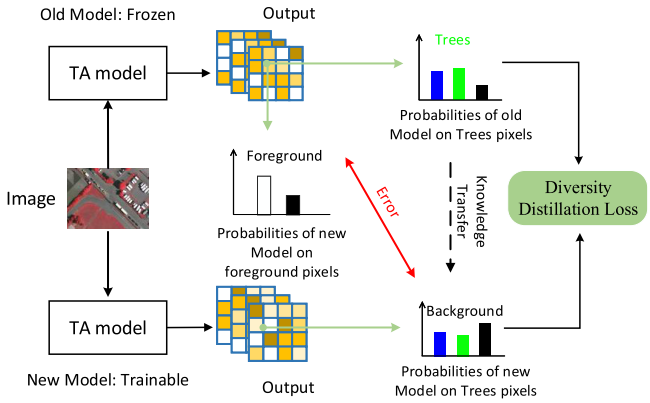


Fig. 7. Illustration of diversity distillation loss. When the binary classifier judges that it is the foreground, but the result of the segmentation classifier of the new task is predicted to be the background, the misjudgment position is distilled.

On the other hand, in order to learn the knowledge of the new tasks, the classification loss is applied here.

1) Classification Loss: When a new task arrives, the segmentation classifier of the new task uses GT for supervision. It should be noted that the model only has the GT of the classes to be segmented in the current task, and the position labels of pixels that do not belong to the current task are recorded as background. So, the loss function for current semantic segmentation during the tth step is defined as

$$L_{\mathrm{ce}}(\theta_t) = \frac{1}{N_t} \sum_{(x,y)\in D^t} l_{\mathrm{ce}}(\theta_t; x, y) \tag{2}$$

where l_{ce} means cross entropy loss in semantic segmentation. The standard cross-entropy loss is defined as

$$l_{\mathrm{ce}}(\theta_t; x, y) = -\frac{1}{|x|} \sum_{u\in x} \log q_x^t(u, y_u) \tag{3}$$

where y_u is the GT of pixel u and $q_x^t(u, y_u) = f_{\theta_t}(x)[u, c]$.

In addition, the binary classifier of every auxiliary module predicts whether each pixel of the input image is foreground or

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

background during the tth step. Among them, the foreground refers to the prediction of the current pixel that belongs to the current task. Therefore, the binary classifier only needs to record the foreground and background as 0 and 1, respectively. We denote the GT mask for binary classifier that only contains the foreground and background as $y_{\text{binary}} \in \{0, 1\}$. So, the binary loss function for the binary classifier is defined as

$$L_{\text{binary}}(\theta_t) = \frac{1}{N_t} \sum_{x \in D^t, y_{\text{binary}}} l_{\text{ce}}(\theta_t; x, y_{\text{binary}}). \tag{4}$$

The segmentation loss function and the binary loss function ensure that the TA model learns the newly added data for the current task. But because the old data is not used for the current model, the performance of the model on the old task will be significantly dropped. The previous literature shows that using distillation loss to constrain the output probability of the old task has a good performance. However, when applied to remote sensing images for semantic segmentation, we found that distillation of the output probability has no advantage on small objects and the edge of objects. Therefore, we propose a diversity distillation loss function to make up for this deficiency.

2) Distillation Loss: The most important problem that CIL should solve is the catastrophic forgetting problem, which ensures that the performance of the old task does not seriously drop. Ideally, incremental learning can be achieved without using any old data so that the TA model learns new tasks without forgetting the old tasks. Knowledge distillation [20] are widely used methods to retain the performance of the old categories. The standard distillation loss is defined as

$$L_{\text{kd}}(\theta_t) = \frac{1}{N_t} \sum_{x \in D^t, f_{\theta_{t-1}}(x)} l_{\text{KL}}[x, f_{\theta_{t-1}}(x)]. \tag{5}$$

Let us take the model trained in the $t-1$th step as the teacher network, and the output of the two classifiers for each task are recorded as $\{\hat{y}^k_{t-1}\}^{t-1}_{k=1}$ and $\{(\hat{y}_b)^k_{t-1}\}^{t-1}_{k=1}$. In order to maintain the performance of the old data, the knowledge of the teacher model needs to be transferred to the student model. The parameters of the teacher model are frozen, and the input images of the teacher network are the same as the student model. Ideally, the student model should have the same distribution of output probabilities as the teacher model. Thus, the common distillation loss is defined as

$$\begin{aligned} L_{\text{dist}}(\theta_t) = & \frac{1}{N_t} \sum_{x \in D^t, \hat{y}^k_{t-1}} l_{\text{KL}}\left(\theta_t; x, \hat{y}^k_{t-1}\right) \\ & + \frac{1}{N_t} \sum_{x \in D^t, (\hat{y}_b)^k_{t-1}} l_{\text{KL}}\left(\theta_t; x, (\hat{y}_b)^k_{t-1}\right) \end{aligned} \tag{6}$$

where l_{KL} denotes as Kullback–Leibler (KL) divergence loss function.

In the classification task, the label of each image contains fixed classes. So, there is no need to consider the relationship between pixels within the image. But in the segmentation task, the goal is to perform label predictions for all pixels on the image. So it is very important to consider the relationship between the foreground and the background, as well as the pixel points between the old task and the new task. Therefore, we design a diversity distillation loss function based on the standard distillation loss function. Through experiments, it can be found that after training the new task, the standard distillation loss function can constrain part of the pixels of the old task. The difference lies in small-scale objects and the edges of objects.

Considering that the binary classification problem has lower complexity, we use a binary classifier to segment the foreground and background. The binary classifier has a higher accuracy for judging the foreground position than the segmentation classifier. Theoretically, for the two classifiers of each task, if the position segmentation classifier is judged as foreground but the binary classifier is judged as background, the position should be focused on. The reverse is also the same. Specifically, when the segmentation classifier predicts a small target or any indistinguishable position as the background, the binary classifier will locate the position and make the model focus on this part of the area. Correspondingly, the diversity distillation loss will not pay special attention to the positions that have been correctly distinguished. Therefore, the diversity distillation loss is defined as

$$L_{\text{dist}-\text{diversity}}(\theta_t) = \frac{1}{N_t} \sum_{x \in D^t, \hat{y}^k_{t-1}} l_{\text{KL}}\left(\theta_t; x, \hat{y}^k_{t-1}\right) * (\hat{y}_b)^k_{t-1}. \tag{7}$$

There are differences between remote sensing images and natural scene images. It is not enough to use the standard distillation loss function to constrain the target in the remote sensing images. Therefore, we propose a diversity distillation loss function for additional constraints, which can better correct the misclassification of small-scale objects and edge pixels of objects. This is of great significance for the incremental learning of the remote sensing image semantic segmentation problem.

D. Conflict Reduction

Incremental learning problem can be defined as two different evaluation settings [12]: 1) single-head (single-classifier) and 2) multihead (multiclassifier).

The single-head evaluation needs to predict all the known labels using one classifier, which means that the classifier should classify all the labels that have been learned. The multihead evaluation refers to the model that uses multiple classifiers to make predictions. For CIL methods, each classifier output the foreground and the background of the corresponding tasks. When applying incremental learning to semantic segmentation, multihead evaluation can avoid the semantic distribution shift of background class and can achieve an unlimited number of tasks to increase. So, our incremental learning model has multiple classifiers. Unfortunately, due to the lack of task ID, each pixel location may be classified into foreground categories at the same time. In order to realize the model predicts and outputs all categories of an image at a time, that is, to achieve single-head prediction. We propose a pseudo-single-head test method.

It should be noted that when the model has multiple classifiers, the probabilities in each classifier are calculated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Process of Conflict Reduction

Input: The size of image is H×W; $f_\theta^k[u]_{seg}$: The probability of the kth segmentation classifier for pixel u of image x during the tth step; $f_\theta^k[u]_{\text{binary}}$: The probability of the kth binary classifier for pixel u of image x during the tth step; q_t^k: the maximum output probability of the kth segmentation classifier of the tth step; $C_{q_t^k}$: the category corresponding to the maximum prediction probability of the kth segmentation classifier of the tth step.

Output: Y': The prediction label of each pixel of the test dataset image; Y: The prediction label of one image in the test dataset.

Establish the prediction label G_t^k of kth segmentation classifier during the tth step.
Establish the prediction label B_t^k of kth binary classifier during the tth step.
while Y in test dataset **do**
  for each pixel u in H × W **do**
    if all B_t^k =Background **then**
      Y' = Background
    else
      if one of the B_t^k = Foreground **then**
        Y' = $G_t^k|_{k=k'}$
      else
        if some of the B_t^k = Foreground **then**
          Y' = $C_{\max}(q_t^k|_{k=k_1,\ldots,k_n})$
        else
          Y'= $C_{\max}(q_k^t|k=1,\ldots,n)$
        end if
      end if
    end if
    return Y'
  end for
  Establish Y according to Y'
  return Y
end while

separately. Different classifiers output different foreground classes. So the chances are that the same pixel in the test image is simultaneously predicted as foreground in different classifiers, which refers to conflict, as shown in Fig. 8.

The purpose of the conflict reduction is to realize that the same pixel can only be predicted as one category realizing the transformation from a multihead incremental learning model to a single-head incremental learning model. It is a general conflict reduction strategy.

Because the prediction result of the binary classifier has higher accuracy, the situation of dividing the same pixel into multiple foreground categories should be reduced. Through experiments, we found that the prediction result of the binary classifier is 1/3 less conflicting pixels than the prediction result of the segmentation classifier. Therefore, we design a fusion mechanism according to the binary classifier.

First, if the prediction label of each binary classifier is the background, the output of prediction is the background. Second, if the prediction label of one binary classifiers

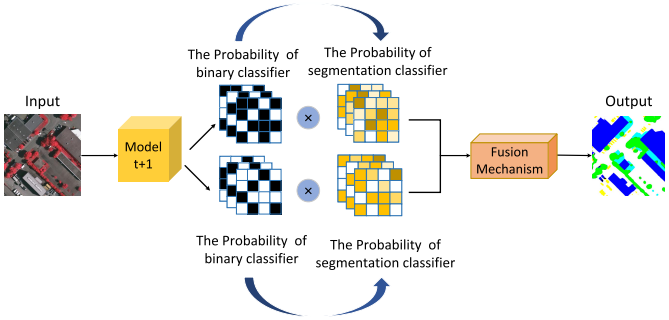


Fig. 8. Schematic of the occurrence of the conflict problem and the location to resolve the conflict problem during testing.

is foreground, the final prediction label is the foreground corresponding to the segmentation classifier. Third, if the prediction label of multiple binary classifier is foreground, the output probability of all channels of the segmentation classifier corresponding to the binary classifier predicted as the background is set to 0. The maximum probability of the segmentation classifier corresponding to the remaining binary classifier predicted as the foreground is the prediction label. Finally, if the prediction label of all binary classifiers is foreground, the category with the highest prediction probability among all segmentation classifiers is used as the final prediction label. At this time, the fusion mechanism is defined as

$$Y' = \begin{cases} \text{Bg} & \text{if all of } B_t^k = \text{Bg} \\ G_t^k|_{k=k'} & \text{if one of } B_t^k = \text{Fg} \\ C_{(q_t^k|_{k=k_1,k_2,\ldots,k_n})_{\max}} & \text{if some of } B_t^k = \text{Fg} \\ C_{(q_t^k|_{k=1}^t)_{\max}} & \text{if all of } B_t^k = \text{Fg} \end{cases} \tag{8}$$

where Y' denotes the predicted label of pixel, B_t^k and G_t^k denote the prediction label of kth binary classifier and segmentation classifier for the tth step, q_t^k represents the maximum output probability of the kth segmentation classifier of the tth step and $C_{q_t^k}$ represents the category corresponding to the maximum predicting probability of the kth segmentation classifier of the tth step.

E. Total Loss

According to the above description, the TA model can simultaneously avoid catastrophic forgetting and learn new classes. When training the network in the tth step, (9) can be formulated as follows:

$$\begin{aligned} L_{\text{total}}^t = {} & L_{\text{ce}}(\theta_t) + L_{\text{binary}}(\theta_t) \\ & + \lambda_1 L_{\text{dist}}(\theta_t) + \lambda_2 L_{\text{dist-diversity}}(\theta_t) \end{aligned} \tag{9}$$

where λ_1 and λ_2 denote two loss weights to balance the total loss function.

IV. Experiments

First, this chapter introduces the dataset and the evaluation metrics used in the experiments. Then we introduced the experimental details and comparison methods. After that, the quantitative analysis results of our method on the Vaihingen

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

and Potsdam datasets are introduced in detail. In the end, qualitative analysis outputs are included to prove the effectiveness of the method.

A. Datasets

We use the Vaihingen and Potsdam datasets with boundaries provided by ISPRS to carry out experiments. The Vaihingen dataset is corrected from the city of Vaihingen in Germany and contains 33 image tiles of different sizes. The training set contains 16 images, and the test set contains 17 images, which is the benchmark specified by the organizer. The resolution of the images is 9 cm. Since the summer of 2018, all images in the training set and test set are public. So we carry out experiments according to the official training set and test set. Besides, we carry out the experiments on the Potsdam dataset that is obtained from Potsdam in Germany. The Potsdam dataset contains 38 images of the same size. The size of each image is 6000×6000 pixels. The resolution of each image is 5 cm. We use the benchmark organizer defined 24 images for training and 14 images for the test following the previous works [58]–[62]. The image contains five channels, namely: near infrared (NIR), R, G, B, and standard digital surface model (DSM) data. Both datasets contain six classes, namely: impervious surfaces (imper surfaces), building, low vegetation (low veg), high vegetation (high veg), or trees, car, and clutter. But since clutter accounts for only 0.78% of the pixels in the dataset, we ignored this class in our experiments, which is also consistent with our baseline (ILSS).

B. Evaluation Metrics

In order to fairly measure the performance of each class on the Vaihingen and Potsdam benchmarks, mean intersection over union (mIoU) [63] and mean F1-score (mF1) [64] score are used as evaluation metrics. The mIoU denotes the average of intersection over union (IoU) of all classes after the tth task. Then mIoU is defined as

$$\text{mIoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{10}$$

where TP, TN, FP, and FN are the number of true positives, true negatives, false positives, and false negatives, respectively.

mF1 score is defined as

$$\text{mF1} = (1 + \beta^2)\frac{\text{precision}}{\beta^2\text{precision} + \text{recall}} \tag{11}$$

where β is the equivalent factor between precision and recall. β is set as 1. In addition, precision and recall are defined as

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{12}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{13}$$

C. Implementation Details

We use PSPNet [50] architecture with ResNet-50 [57] pretrained on ImageNet [65] as the TA model' main body. The classifier includes a 1×1 convolution and a softmax operation. During the first step, the learning rate is $1e - 2$, and the learning rate is set to $4e - 3$ for all the following ones. During training, the momentum and weight decay are set to 0.9 and 0.001. The TA model is trained with stochastic gradient descent (SGD) optimizer. We use the poly strategy to update the learning rate for each iteration. All baselines keep the same settings. We randomly crop each large image into 500 small images of 513×513 pixels on the Vaihingen dataset and the Potsdam benchmarks. The training dataset on the Vaihingen dataset and Potsdam dataset contain 8000 and 12 000 small images, respectively. We train the model for 30 epochs for the Vaihingen dataset and ten epochs for the Potsdam dataset. For all methods, we apply the same data augmentation, including random horizontal flip, random rotation, and random Gaussian blur. During testing, we use a sliding window strategy with overlapping stride $1/3$ to verify performance on a single scale and original inputs. All experiments are conducted by Pytorch [66] on the NVIDIA TITAN RTX GPU.

D. Compared Methods

In order to evaluate the effectiveness of our method, compared methods include four common methods and four latest methods. Among them, four latest methods include ILT [19], ILSS [7], ModelIng the Background (MiB) [34], and EWC [2]. Four common methods include SL, fixed-representation (FR), FT, and FT (PPM).

1) Static Learning: Compared with incremental learning methods, all the training data, as well as GT, can be accessed at once in the training stage in SL. SL can be regarded as an upper bound for incremental learning.

2) Fixed-Representation: FR method fixes the feature extractor of the model after learning the first task and adds a classifier that is the same as the old task. The new classifier is initialized randomly. Only the GT of the classes in the new task is used during training, and the annotations of the classes in the old task cannot be accessed. FR does not perform well on the new task because the feature extractor cannot learn the features of the new data. Still, it can avoid the catastrophic forgetting problem to a certain extent.

3) Fine-Tuning: The model structure and input data of FT and FR are the same. The only difference is that FT does not freeze the feature extractor and trains all parameters for the current model. FT can learn new knowledge, but the model will forget the knowledge of the old tasks. Therefore, the model will encounter catastrophic forgetting on old tasks.

4) Fine-Tuning (PPM): The only difference between FT (PPM) and FT is the model structure. When new tasks arrive, FT (PPM) adds a PPM module.

5) Incremental Learning Techniques [19]: As the first incremental learning paper applied to semantic segmentation, ILT is a commonly used compared method. When training a new task, the data of old tasks is also inaccessible. The article proposes four distillation strategies to preserve the knowledge of the old task. We use the method of distilling the logits, which has better accuracy. While learning new tasks, the catastrophic forgetting problem of the model is improved.

6) Elastic Weight Consolidation [2]: EWC remembers old tasks by selectively slowing down learning on the weights important. EWC applies the Bayesian framework to neural networks, which allows determining the posterior distribution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

TABLE I
ABLATION STUDY FOR EACH COMPONENT

Methods	L_{dist}	$L_{dist-diversity}$	CR	Task1 Building	Task1 High veg	Task1 Imper surf	Task2 Low veg	Task2 Cars	mF1(%)
TA Net	✓			92.02	85.34	89.26	79.46	72.93	83.80
	✓	✓		93.4	86.09	89.52	79.54	77.56	85.22
	✓	✓	✓	93.41	86.15	89.54	79.66	77.66	85.28

TABLE II
ABLATION STUDY FOR CONFLICT REDUCTION

Methods	Building	High veg	Imper surf	Low veg	Cars	mF1(%)
noCR	91.59	81.9	88.47	74.48	52.36	77.76
logits_womask	93.4	86.07	89.34	78.75	75.63	84.64
CR_withmask	93.41	86.15	89.54	79.66	77.66	85.28

TABLE III
PREDICTION RESULTS OF THE BINARY CLASSIFIER AND SEGMENTATION CLASSIFIER UNDER DIFFERENT TASKS

Task Setting	binary-classifier mF1	binary-classifier mIoU	segmentation-classifier mF1	segmentation-classifier mIoU
2-3	92.74	86.46	92.01	85.87
3-2	93.98	88.65	89.74	81.52
4-1	99.47	98.94	87.32	77.82

TABLE IV
NUMBER OF CONFLICT PIXELS ON VAIHINGEN TEST SET

Methods	binary-classifier	segmentation-classifier
Total number of conflict points	1345282	1979112
same as foreground	904237	1470349
same as background	441045	508763

TABLE V
ABLATION STUDY FOR λ_1 AND λ_2. IN THE EXPERIMENT, λ_1 IS FIXED AND λ_2 IS CHANGED

λ_1	λ_2	mF1(%)	mIoU(%)
10	1	84.82	74.26
10	5	84.98	74.31
10	10	85.28	**74.82**
10	15	85.18	74.65
10	20	84.91	74.35

TABLE VI
ABLATION STUDY FOR λ_1 AND λ_2. IN THE EXPERIMENT, λ_2 IS FIXED AND λ_1 IS CHANGED

λ_1	λ_2	mF1(%)	mIoU(%)
1	10	85.12	74.55
5	10	85.32	74.81
10	10	85.28	**74.82**
15	10	85.05	74.39
20	10	84.91	74.35

of parameters by introducing uncertainty on the network parameters, not just the point estimation in the parameter space.

7) Modeling the Background [34]: Aiming at the problem of semantic distribution shift of background class, modeling the background (MIB) proposes to reconstruct the dark knowledge of student network and teacher network, which alleviates the catastrophic forgetting problem of old tasks.

8) Incremental Learning for Semantic Segmentation [7]: ILSS, as the first incremental learning article on semantic segmentation using remote sensing datasets, is one of the most important baselines in our article. ILSS uses the traditional knowledge distillation method to retain the information of the old tasks and combines the binary cross-entropy loss to predict the segmentation label of each pixel. However, this method did not consider the characteristics of remote sensing images. When only small-scale objects are included in the foreground, there are too few foreground pixels, which will cause the problem of unbalanced categories. Besides, due to the expansion and contraction of edge positions, the catastrophic forgetting problem is further deepened.

E. Experiments on Vaihingen Dataset

When experimenting on the Vaihingen dataset, we keep the same experimental setting as ILSS. The number of categories for each task is $2-2-1$. In addition, we also use the following order for the increase: $4-1, 3-2, 2-3$. During the experiments, we assume that the data of the old task is not accessible, and the unknown pixels are set as the background class.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
PERFORMANCE COMPARISONS ON THE VAIHINGEN DATASET FOR $4-1$ CIL SCENARIOS. BEST IN **BOLD**, RUNNER-UP *Italics*

Methods	Task1				Task2	mF1(%)	mIoU(%)
	Building	High veg	Imper surf	Low veg	Cars		
Fine-Tuning	47.06	3.28	8.38	0.76	67.34	25.36	17.59
Fine-Tuning(PPM)	20.87	9.36	21.51	0.54	61.42	22.74	16.25
Fixed Representation	***93.25***	**86.07**	***89.43***	***80.31***	1.8	70.17	62.33
EWC	0.21	36.52	6.69	27.14	78.69	29.85	21.12
ILSS	86.06	83.37	82.9	71.47	6.1	65.98	55.32
MIB	92.43	85.27	88.61	78.44	76.25	***84.2***	***73.96***
ILT	91.38	85.08	87.54	78.17	***77.71***	83.98	72.74
Ours	**93.38**	***86.03***	**89.71**	**80.66**	**79.92**	**85.94**	**75.73**
Static-Learning	93.15	86.24	89.46	80.26	76.86	85.19	74.67

1) Effect of Each Component: In order to verify the role of each part of our method, we carry out ablation experiments for the $3-2$ experimental setting, as shown in Table I. First, the model of task1 trains the three categories of Building, High veg, and Imper surf. In order to retain the knowledge learned in task1, the output of the teacher model is distilled using the common distillation loss proposed by Hinton [20]. The final mF1 score is 83.80. It can be seen that the knowledge learned in task1 encounters the problem of catastrophic forgetting after the model learns the knowledge of the new task. In the second row of the table, we add the diversity distillation loss to train the model. During testing, the model uses multihead evaluation. Each classifier only needs to predict the category of an old task during multihead prediction. Experiments show that adding the diversity distillation loss function can realize that the knowledge of the old task is effectively retained under the premise that the accuracy of the new task is not reduced so that the catastrophic forgetting problem is better alleviated. In response to the conflict problem of multihead prediction, we designed a conflict reduction mechanism to achieve single-head output for the entire picture without reducing the accuracy. The result is shown in the third row of Table I. The conflict reduction mechanism allows the multihead classifier to output all categories of the entire image at one time, without relying on the prior information of the task-ID.

2) Effect of Binary Classifier: In order to verify the effectiveness of our method in this article, we compare the prediction results of the binary classifier and segmentation classifier. The experiment was carried out under three settings of $2-3$, $3-2$, and $4-1$, respectively. The experimental results are shown in Table III, which shows the mIoU and mF1 score of foreground prediction results by the binary classifier and segmentation classifier. In a horizontal comparison, the mIoU and mF1 scores of the binary classifier are higher than the mIoU and mF1 scores of the segmentation classifier. From a longitudinal perspective, the more categories learned in the first task, the higher the mIoU and mF1 scores of the binary classifier. The result of the segmentation classifier decreases as the categories learned by the task increase. We consider that the binary classification task is simpler than the task of segmentation classification task, so when training the data of the new task, the result of the binary classifier is used to supplement the distillation loss. The knowledge distillation methods have the potential to retain the performance of the old task. But we found that there is still a gap in accuracy between the old model and the new model. Therefore, we emphatically constrain the hard cases that are still misclassified in order to alleviate catastrophic forgetting.

3) Ablation Study for Conflict Reduction: When testing the existing CIL model, it can be roughly divided into two types: single-head testing and multihead testing. When the incremental learning model is designed for the semantic segmentation datasets, because the single-head incremental learning model will encounter the background distribution shift problem during the test, the TA model adopts the multihead mode. However, on the one hand, the prediction output of the model for the same pixel is in conflict during testing. The model tends to classify the same pixel into a category with higher prediction accuracy for this classification head, and the accuracy at this time may be higher than the single-head output. On the other hand, we think that the predicted output of each classification head is the background class and still has certain information waiting for our mining. Therefore, we expect to design a universal conflict reduction mechanism that will not reduce the accuracy. All multihead incremental learning models can perform single-head predictions without loss of accuracy.

In the experiment, we take the $3-2$ task setting as an example. The first task trains three classes, namely building, high vegetation, and impervious surface, and the second task trains two classes, namely low vegetation and cars. In order to meet the single-head test, we designed three methods, and the results are shown in Table II. First, the segmentation classifier of the first task predicts the foreground as building, high vegetation, and impervious surface, and the other classes are set as the background. The segmentation classifier of the second task predicts the pixels where low vegetation and cars are located as foreground. At this time, the pixel positions predicted by the classifiers of all tasks as foreground should be complementary in an ideal situation. Specifically, we first perform softmax operation on the output feature map. Then, we concatenate the channels of different tasks and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII

PERFORMANCE COMPARISONS ON THE VAIHINGEN DATASET FOR 2 – 3 CIL SCENARIOS. BEST IN **BOLD**, RUNNER-UP ***Italics***

Methods	Task1		Task2			mF1(%)	mIoU(%)
	Building	High veg	Imper surf	Low veg	Cars		
Fine-Tuning	67.98	18.69	***89.72***	**80.66**	***77.09***	66.83	55.2
Fine-Tuning(PPM)	58.16	21.39	**89.73**	80.52	**79.02**	65.76	53.41
Fixed Representation	**93.92**	**86.33**	33.18	26.95	0.37	48.15	39.45
EWC	77.34	73.32	89.15	***80.45***	73.58	78.77	64.95
ILSS	90.29	85.2	87.87	79.41	40.82	76.72	65.27
MIB	92.23	***86.31***	89.18	80.1	72.96	***84.16***	***73.72***
ILT	91.78	84.23	88.54	80.19	72.26	83.4	73.45
Ours	***93.6***	86.25	89.34	80.4	74.09	**84.74**	**74.38**
Static-Learning	93.15	86.24	89.46	80.26	76.86	85.19	74.67

TABLE IX

PERFORMANCE COMPARISONS ON THE VAIHINGEN DATASET FOR 2 – 2 – 1 CIL SCENARIOS. BEST IN **BOLD**, RUNNER-UP UNDERLINED

Methods	Task1		Task2		Task3	mF1(%)	mIoU(%)
	Building	High veg	Imper surf	Low veg	Cars		
Fine-Tuning	1.23	0.17	6.57	3.1	72.14	16.64	10.49
Fine-Tuning(PPM)	9.9	32.75	3.62	38.87	69.2	30.87	21.72
Fixed Representation	**93.71**	***86.33***	46.37	34.75	1.97	26.23	19.21
EWC	66.93	52.59	62.81	70.33	73.61	65.25	53.89
MIB	92.38	84.99	***88.24***	***78.46***	74.52	***83.72***	***72.78***
ILSS	87.41	83.2	81.48	72.25	4.32	65.73	54.25
ILT	91.24	84.46	88.19	77.2	***74.66***	83.15	72.35
Ours	***93.12***	**85.73**	**89.32**	**79.49**	**75.74**	**84.68**	**73.96**
Static Learning	93.15	86.24	89.46	80.26	76.86	85.19	74.67

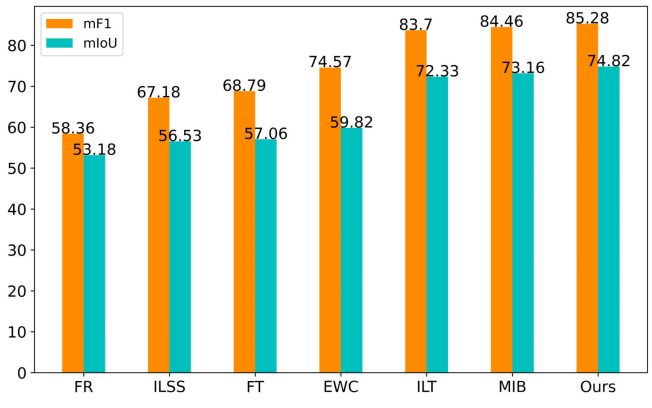


Fig. 9. Under 3 – 2 task settings, the mF1 scores and mIoU of all methods on the Vaihingen dataset.

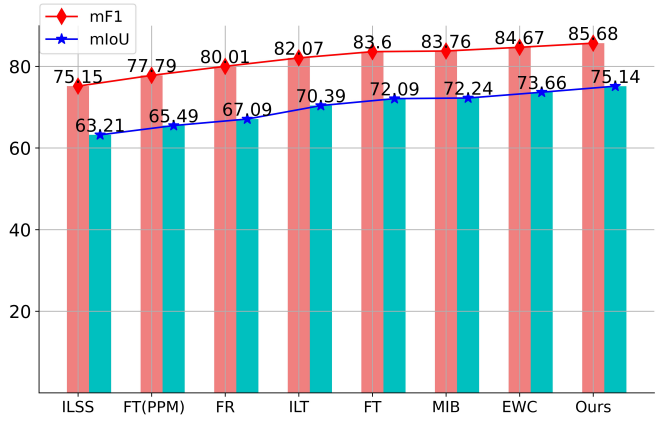


Fig. 10. Under 2 – 3 task settings, the mF1 and mIoU values of all methods on the Potsdam dataset.

then perform the argmax operation by channel. The index corresponding to the maximum value is the predicted category of each location. This is the simplest way to design conflict reduction, which is denoted as "noCR" in Table II. However, the experimental results showed that the ideal situation did not appear, and the background of all tasks had conflicting points. There are two reasons for this situation. On the one hand, different classifiers may predict the foreground category at the same time. On the other hand, the probability value predicted by one classifier as the background may be greater than the probability value predicted by the other classifier as the foreground. Therefore, it is unreasonable to concatenate according to the channel.

Therefore, in order to avoid the foreground and background pixel conflicts of prediction label by multiple classifiers, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

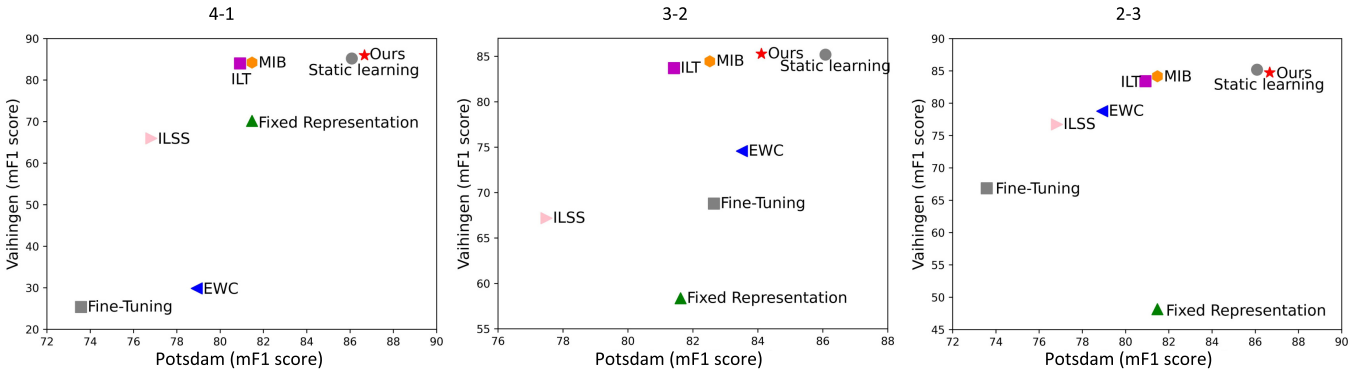


Fig. 11. mF1 score on the Vaihingen and Potsdam dataset under the same experimental settings. Our experiments have the best effect among all incremental learning methods and have the same effect as the SL. In the $4-1$ experimental setting, it even surpasses the SL.

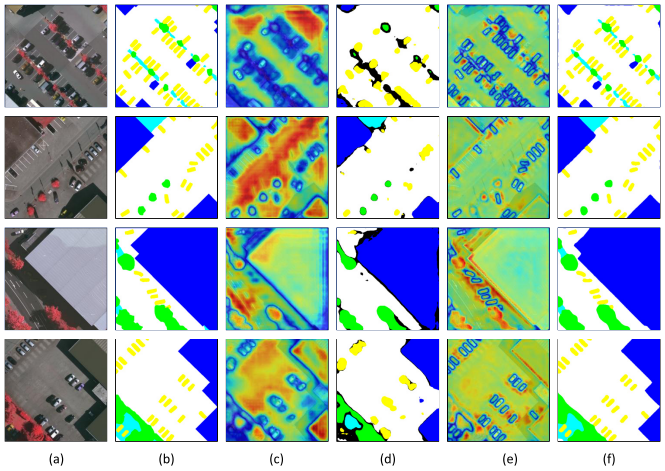


Fig. 12. Visualization of small-scale targets. (a) Input image. (b) GT. (c) Heat map of baseline (ILSS). (d) Output of baseline (ILSS). (e) Heat map of ours. (f) Output of ours.

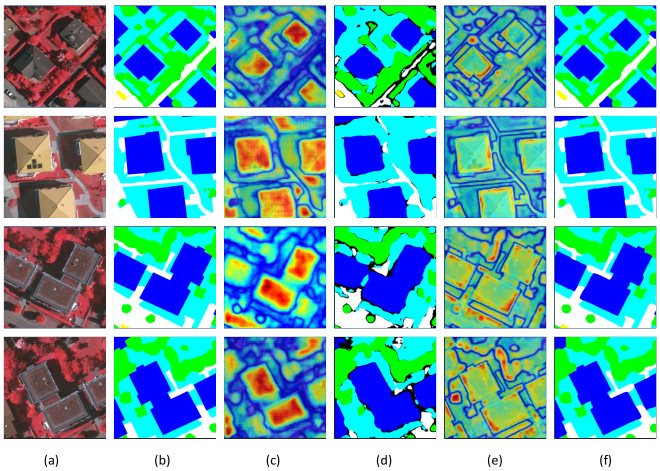


Fig. 13. Visualization of edge. (a) Input image. (b) GT. (c) Heat map of baseline (ILSS). (d) Output of baseline (ILSS). (e) Heat map of ours. (f) Output of ours.

second conflict reduction method named "logits-womask" in Table II. If the predicted label of the classifier is background, the probability value of all channels of the classifier will be set to $1e-6$. In this way, the classifier whose prediction result is background will not affect the classification whose prediction result is foreground. When the prediction results of two or more classifiers are foreground at the same time, we take the index corresponding to the maximum value of all the probabilities as the prediction output result. If the prediction results of all the classifiers are background, the predicted label of this pixel is background. At this time, the mF1 score increased by 6.88 points.

Since the binary classifier is more accurate in distinguishing the foreground and the background, in the third method, we first carry out an experiment on the results of the binary classifier. The experimental results are shown in Table IV. The number of pixels of the binary classifiers that simultaneously judged as foreground or background by multiple classification heads is $1/3$ less than that of the segmentation classifier, which is about 60w pixels. Therefore, using the foreground of the binary-classifier to guide the segmentation classifier will have better performance. The experimental results show that this method increases the value of mF1score by 0.64 compared with the second conflict reduction strategy.

4) Ablation Study for Multiple Loss Functions: The TA model utilizes multiple loss functions to optimize the learning process. Each parameter will have an impact on the final result. λ_1 and λ_2 are the weight parameters of common distillation loss and diversity distillation loss, respectively. The experimental results for λ_1 and λ_2 are presented in Tables V and VI. We fix λ_1 equal to 10, and the values of λ_2 are equal to 1, 5, 10, 15, and 20, respectively. In this state, the performance of the model has increased first and then decreased. When λ_2 is equal to 10, the model has the highest mIoU during testing. The same applies when determining the value of λ_1. The model achieved the best mIoU when λ_1 and λ_2 were both equal to 10. The mF1 score of the model tested on Vaihingen data is 85.28.

The purpose of this ablation experiment is to verify the influence of common distillation loss functions and diversity distillation loss on the results when the model is training. The experimental results show that it has the best performance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

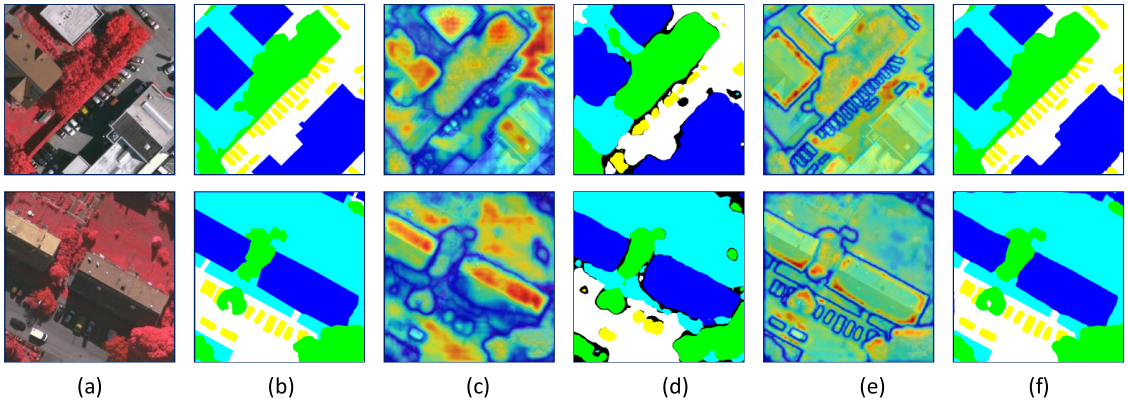


Fig. 14. Visualization of edge and car. (a) Input image. (b) GT. (c) Heat map of baseline (ILSS). (d) Output of baseline (ILSS). (e) Heat map of ours. (f) Output of ours.

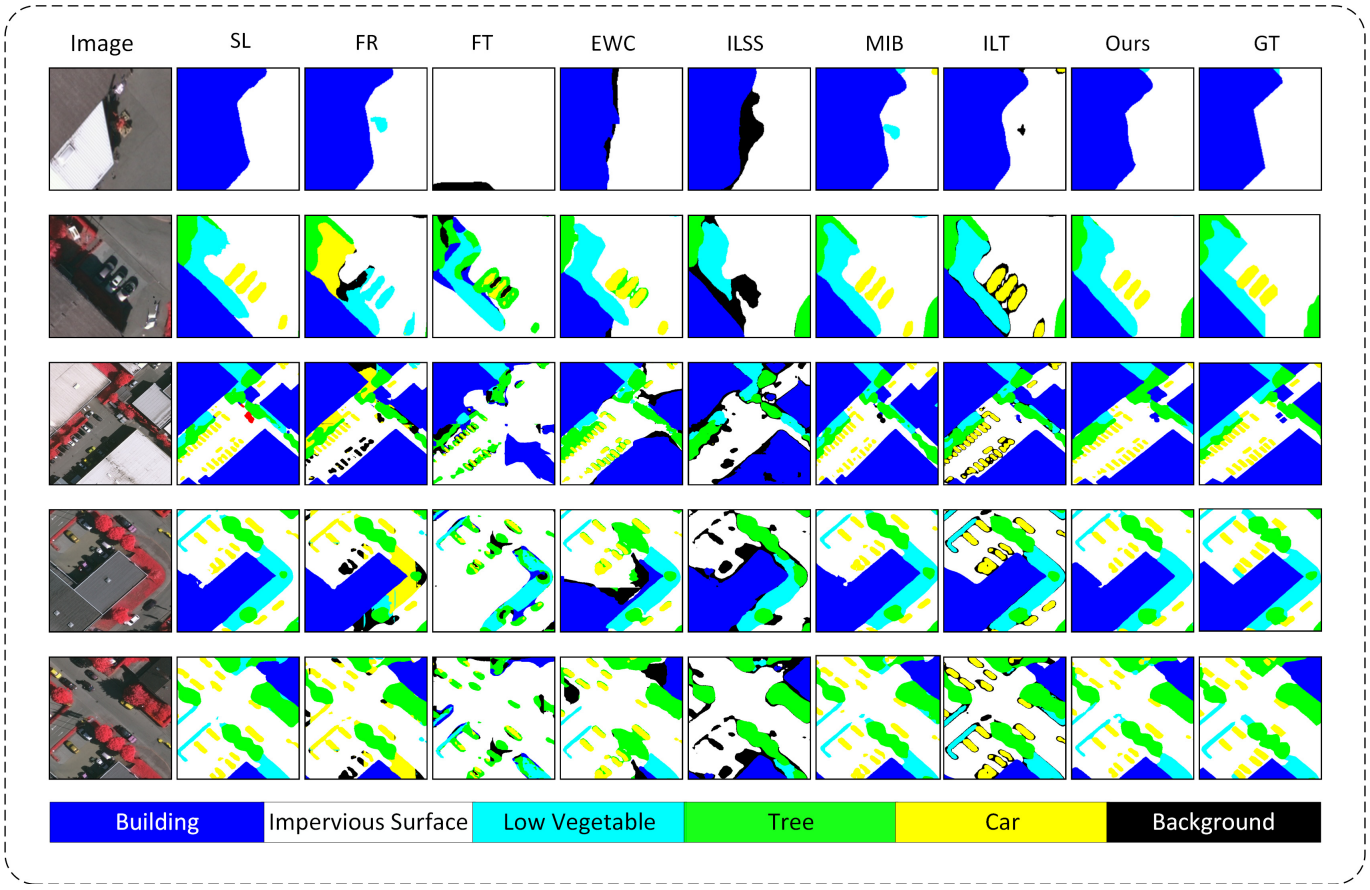


Fig. 15. Visualization of different methods on Vaihingen dataset.

when the common distillation loss and the diversity distillation loss are equal. We consider that common distillation loss has better retention for the segmentation subject. At the same time, our proposed diversity distillation loss has a good effect on the edges and small objects that are difficult to distinguish. The diversity distillation loss provides a good auxiliary effect for the common distillation loss and has a good correction for the label prediction of the pixels that are difficult to distinguish.

5) Comparison With State-of-the-Art: Addition of one class $(4-1)$.

In this experiment, we perform two learning steps: the first in which we observe four classes, and the second where we learn the cars class. The first four classes are building, high vegetation, impervious surfaces, and low vegetation. Results are reported in Table VII. As expected, FT and FT (PPM) perform worse than FR in old classes. They suffer from catastrophic forgetting seriously. As for FR, though the old

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

TABLE X
PERFORMANCE COMPARISONS ON THE POTSDAM DATASET FOR $4-1$ CIL SCENARIOS. BEST IN **BOLD**, RUNNER-UP UNDERLINED

Methods	Task1				Task2	mF1(%)	mIoU(%)
	Building	High veg	Imper surf	Low veg	Cars		
Fine-Tuning	81.63	54.81	76.64	68.89	**85.93**	73.58	59.34
Fine-Tuning(PPM)	39.36	0.05	2.77	0.42	83.24	25.17	19.42
Fixed Representation	90.11	75.84	84.43	78.39	78.56	81.47	68.49
EWC	87.44	76.45	80.46	77.65	72.58	78.92	65.46
ILSS	***93.59***	***82.1***	***86.51***	81.2	40.68	76.82	65.54
MIB	90.83	78.34	84.45	***79.36***	74.36	**81.47**	**69.13**
ILT	88.8	72.08	81.68	77	85.05	80.92	68.37
Ours	**93.82**	**82.75**	**88.89**	**82.42**	***85.46***	**86.67**	**76.73**
Static-Learning	92.92	82.06	87.93	80.37	87.1	86.08	75.28

classes do not be forgotten, the learning of the new class encounter difficulties. ILT applies distillation loss in outputs so as to keep the performance for old classes. This confirms the effectiveness of this approach on preventing catastrophic forgetting. MiB surpasses ILT because it focuses on the importance of MIB for semantic segmentation. Compared with all baseline methods, our method achieves the best performance. The mF1 score of incremental methods is even higher than SL by 0.75% points and slightly higher than multi learning. It can be seen that our method has a significant improvement on the cars. This is in line with our method for the accurate positioning of small-scale objects. When the old model does not misjudge this class, it will have better performance. We can find this phenomenon in the next experiments.

6) Comparison With State-of-the-Art: Addition of two class $(3-2)$.

In the first experiment, we initially train the network on three classes and we add the remaining two classes all at once. The first task contains building, high vegetation and imper surface and the next task contains low vegetation and cars. The results are shown in Fig. 9. The experimental results are analyzed from three aspects. First, from the view of the degree of oblivion to old tasks, the performance of the FR on the old task has a better mF1 score than our method. MIB followed closely. The performance of other methods on the old task is not competitive with ours. Due to the fixed feature extractor in the FR method, the ability to represent old tasks can be maintained. However, the model with a fixed feature extractor is not conducive to the extraction of new task features. Next, from the perspective of learning new tasks, FT and FT (PPM) have relatively good on the new task because the FT did not take methods to retain the knowledge of the old task. FT is a method of transfer learning, which can realize the transfer of knowledge from task A to task B. But maintaining the performance of task A is not a concern for FT. Therefore, the use of FT methods can help the learning of new tasks and can even achieve better results than SL. Our method has similar performance to EWC, except for the FT and FT (PPM), which has the closest result to the SL. Finally, from the view of overall performance, only our method can have the best performance on the new class and the old class at the same time, which also proves the effectiveness of our method. It should be noted that the car still has the best performance in all methods.

7) Comparison With State-of-the-Art: Addition of three classes $(2-3)$.

In this experiment, the first task contains only two classes, namely building and high vegetation. The remaining three classes of labels are will be learned as the second task. The results are shown in Table VIII. When the number of classes in the new task is more than the number of classes in the old task, the network learning encounters greater difficulties. As shown in Table VIII, FR on the old task has a better mF1 score than our method, which is 0.2 points higher than our method on average. However, this method has a relatively lower performance on new tasks. The performance of FR on the new task is about 50 points lower than our method. Compared with FR, our method has the ability to continuously extract the features of new data. At the same time, our method has a competitive effect in retaining knowledge of old tasks. Table VIII shows that MIB has 0.06 points mF1 score slightly higher than our method on the high vegetation. Our method has 1.27 mF1 score higher than MIB on the building category. Our method is superior to MiB method on the old tasks. Besides, our method has 1.15 points than MiB in cars category. According to the performance of the new tasks, MiB is relatively uncompetitive. As shown in Table VIII, FT has a higher mF1 score on new tasks compared with other methods. But FT method suffers from the problem of catastrophic forgetting. The performance on the old task is about 40 points lower than our method. ILT has improved the performance of FT unsurprisingly, but the performance is still not as good as our method. It is worth noting that in this experimental setting, the performance of cars is not better than SL, but it is still superior to other regular-based methods.

Incremental learning focuses on the overall performance on all tasks. The above three methods are better than our method on a certain category. Other baseline methods did not surpass our method on other categories. Although FR has relatively good performance on the old task, its performance on the new task is about 50 points lower than our method. For FT, it has outstanding performance on the new task, but the performance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

TABLE XI

PERFORMANCE COMPARISONS ON THE POTSDAM DATASET FOR 3 − 2 CIL SCENARIOS. BEST IN **BOLD**, RUNNER-UP UNDERLINED

Methods	Task1			Task2		mF1(%)	mIoU(%)
	Building	High veg	Imper surf	Low veg	Cars		
Fine-Tuning	88.68	78.58	85.21	78.6	82.19	82.65	70.61
Fine-Tuning(PPM)	83.96	65.52	80.3	**80.75**	***84.46***	79	65.79
Fixed Representation	90.71	76.57	83.2	77.48	80.13	81.62	69.27
EWC	90.2	78.53	85.33	79.67	83.83	***83.51***	***75.22***
ILSS	**92.7**	**81.46**	***86.96***	***80.61***	45.72	77.49	65.84
MIB	90.55	78.88	82.76	76.23	84.2	82.52	70.55
ILT	89.15	70	82.8	79.76	**85.39**	81.42	69.15
Ours	***91.67***	***79.65***	87.99	77.83	83.39	**84.11**	**75.92**
Static-Learning	92.92	82.06	87.93	80.37	87.1	86.08	75.28

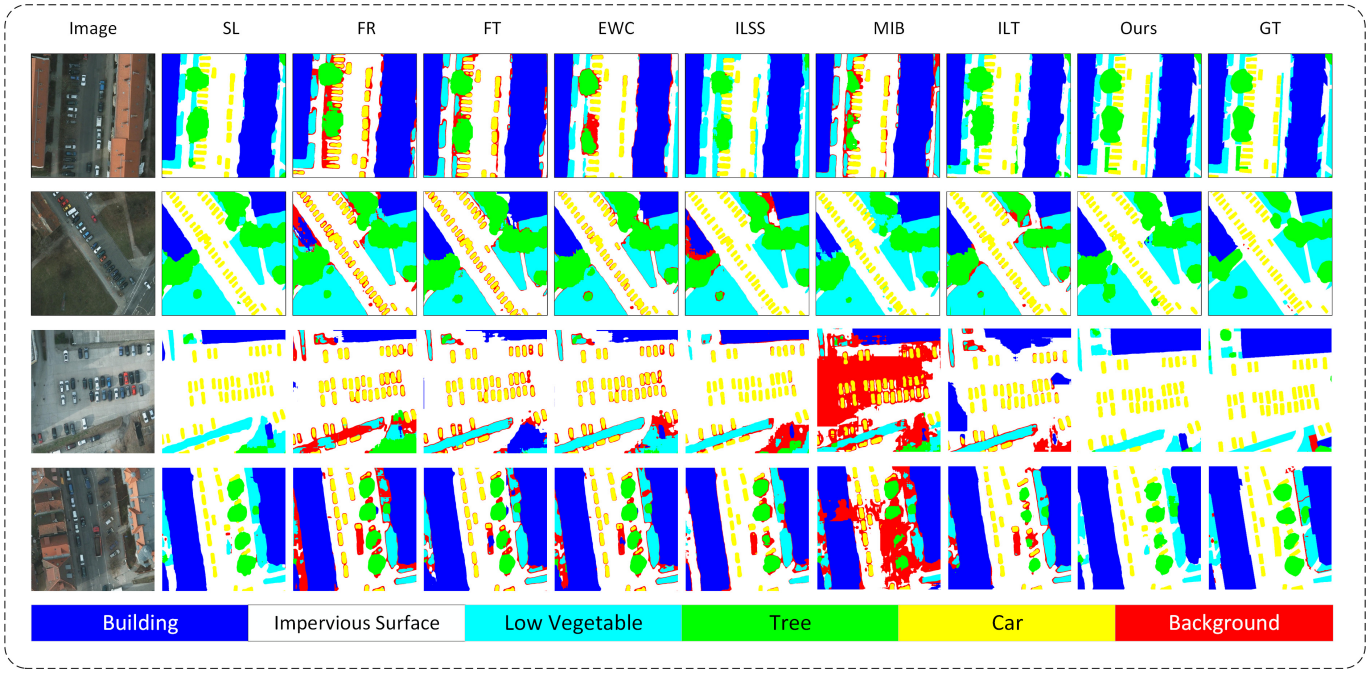


Fig. 16. Visualization of different methods on Potsdam dataset.

on the old task is about 40 points lower than our method. MiB only slightly outperforms our performance on the category of high vegetation. But our method is 1.27 and 1.15 points higher than MiB in the building and cars categories, respectively. Therefore, the performance of our method on the old and new tasks ranks ahead of all comparison methods. Our method has a better performance in retaining the knowledge of the old tasks and learning the knowledge of the new tasks. This also proves the effectiveness of our method for improving the performance of small-scale objects.

8) Comparison With State-of-the-Art: (2 − 2 − 1).

In order to verify the effectiveness of our method on more tasks, we divide the Vaihingen dataset into three tasks, which is consistent with the ILSS. The experimental results are shown in Table IX. As expected, SL outperforms the other methods on the Vaihingen dataset. But it should be noted that the SL uses all annotations, which is not developmental in the iteratively updated era of big data. In FT and FT (PPM) methodology, the performance of task1 and task2 are dropped seriously. FR is the opposite of FT and even has the second best performance on the old task (bold and italic). In terms of new tasks, FT and FT (PPM) has a better effect because network parameters change based on the knowledge of the new task. EWC uses the model parameters trained in the old task to constrain the parameters that are about to change, which has a slight effect in avoiding catastrophic forgetting. ILSS has a similar performance to EWC on mF1 score. But there is a big difference in the performance of the new and old tasks. Specifically, ILSS is more effective in retaining knowledge of old tasks. However, when only learning small target objects, the performance is not good because the number of foreground and background pixels is not balanced. ILT distilled the distribution of output probability, and mF1 score reached 83.15 points. MIB has a better effect than ILT by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

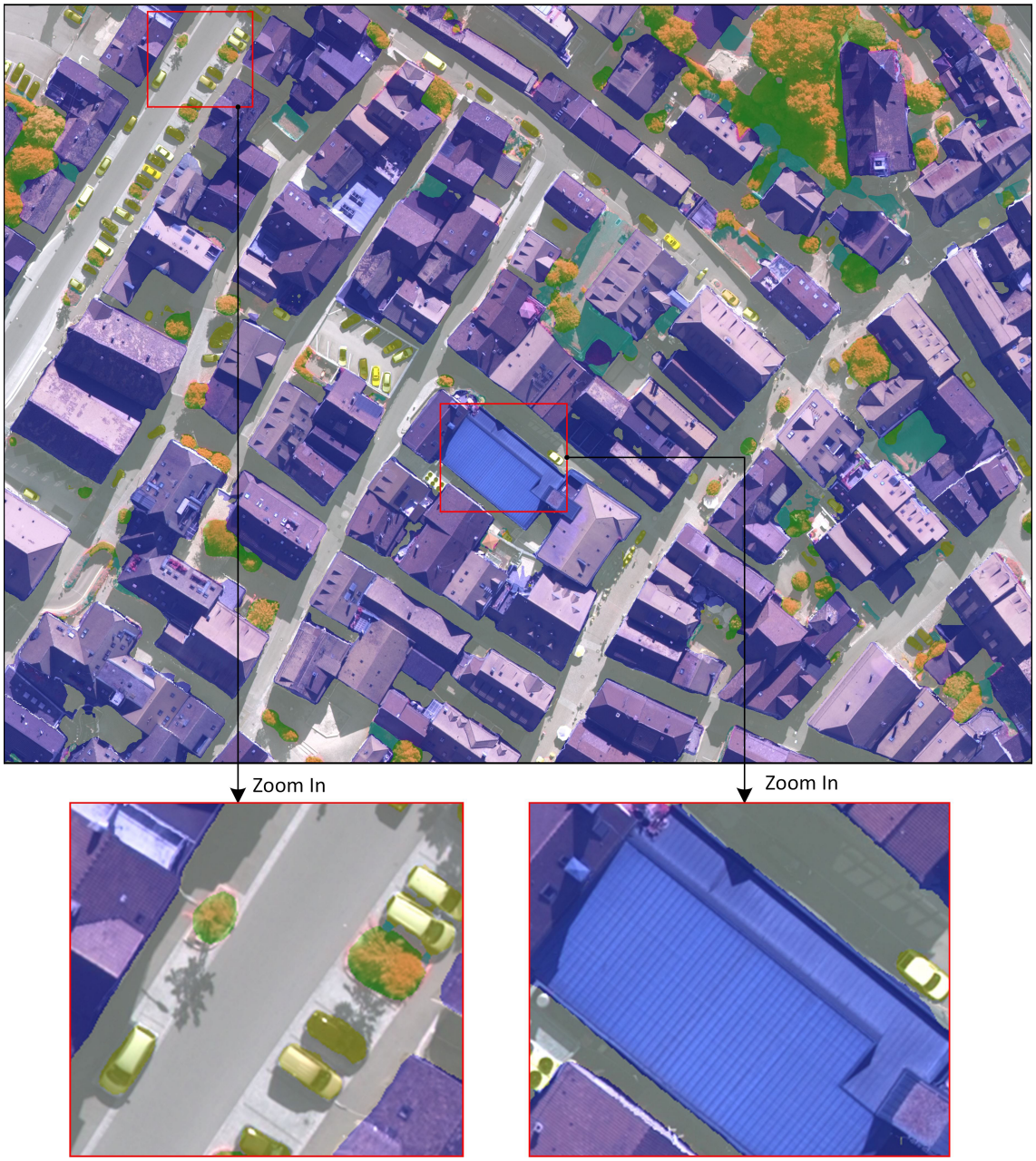


Fig. 17. Original image and its pixel-level classification results with an image size of 1887 × 2557 pixels. We zoomed in the red rectangle that shows the small objects and the edge of the building.

updating its distillation strategy. Our method focuses on the distillation of forgotten pixel positions on the basis of not encountering the semantic shift of the background class, and still has the best performance under the condition of multiple tasks.

F. Experiments on Potsdam Dataset

To further verify the effectiveness of our method, the experiments on Potsdam benchmark by ISPRS are carried out. The experimental setup is the same as the Vaihingen dataset, except that the number of epochs is set to 10. The incremental learning settings in the experiment are $4-1$, $3-2$, and $2-3$, respectively. Numerical comparisons with SOTA methods are shown in Tables X and XI, and Fig. 10. Our method achieves the best mF1 score among all the comparison methods. In the table, we bolded the best-performing of each category, and bolded and italicized the second-performing. In addition, in order to comprehensively measure the performance of our method on the two datasets, we visualize the mF1 score according to each incremental learning setting, as shown in Fig. 11. Our method is superior to all task incremental learning methods and is the result of the closest approximation to the SL. In the $4-1$ experimental setting, our method outperforms SL on both datasets.

G. Qualitative Analysis

We visualized the features to intuitively express the superior performance of our method on boundaries and small targets.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

Since our method has a better performance for small target objects and object edges, we focus on the analysis of these two parts in Sections IV-G1 and IV-G2. In addition, we visually compare our method with the SOTA methods.

1) Visualization at Small-Scale Targets: The visualization result of the heat map of small-scale targets is shown in Fig. 12. The heat map can highlight the important information of the feature map. The red and yellow color in images is more important and blue color means less important. The input images in the first column come from the Vaihingen dataset. The second column shows the GT corresponding to the first column. When applying to the baseline method (ILSS), the heat maps of the feature map are shown as the third column and the outputs are shown as the fourth column. Because there are more small-scale targets in remote sensing datasets than natural scene datasets, their accuracy is particularly important. However, ILSS fails to segment small-scale objects. As shown in Fig. 12(c), the position of the cars is the weakest position that has been paid attention to. Besides, the edge of cars is not clear. The focus of the heat map focuses on the categories of the old tasks, such as the floor, which is in line with the output result. Compared with the baseline method, our method is very effective in distinguishing small targets, which is shown in Fig. 12(e) and (f). Our method has no obvious tendency to focus on the old task and new task. The boundaries of the car are also well divided, which shows that our method is effective in solving the catastrophic forgetting problem caused by small-scale targets.

2) Visualization at the Edge: The visualization results of the heat map at the edge are shown in Fig. 13. In the process of exploring the causes of catastrophic forgetting, we found that edge expansion and contraction of remote sensing data have a greater impact on the performance of the old task. The first column and the second column are input images and the corresponding GT in the Vaihingen dataset. The third column is a heat map by the feature extractor in the baseline method. The heat map in the third column mainly pays attention to building and impervious surface that belong to the old task. But the edges of buildings shown in the heat map have been deformed to varying degrees, which is also reflected in the output result (in the fourth column). After applying our method, the edges of objects, such as buildings are corrected, which is shown in Fig. 13(e) and (f). The model pays more attention to the location near the edge. At locations outside the edge, the model pays almost the same attention to the image, and there is no obvious tendency for the new task and the old task. This proves that our method alleviates the catastrophic forgetting problem caused by the edge. In addition, the visualization result of the heat map at small-scale targets and edges are shown in Fig. 14.

3) Visualization of Segmentation Results: First, we visualize small patches, focusing on edges and small target areas in Fig. 15. Compared with the baseline method, the visualization effect has obvious advantages. Specifically, the visualization result shows that FR avoids the catastrophic forgetting problem. However, it fails to learn new knowledge from new data. On the contrary, FT has forgotten what learned before and has good segmentation performance for new tasks. Compared with FT, EWC performs better on the old tasks and is the same as it on the new task. After that, ILSS is proposed to be applied to remote sensing datasets to achieve incremental learning, which is the first paper for remote sensing images. ILSS does not perform well for edges and small-scale targets, though old knowledge is well preserved. Besides, ILT and MiB also have different degrees of improvement for the catastrophic forgetting problem. Instead, our method is the first article that considers the relationship between remote sensing characteristics and catastrophic forgetting problem, which achieves better results on edges and small target objects. Then, the qualitative results on the Potsdam dataset are presented in Fig. 16. Then, we visualized the entire 1887×2557 pixel image in Fig. 17 and zoomed in on some areas, and we can get similar conclusions.

V. Conclusion

In this article, we study the class-ILSS in aerial imagery. We observe the characteristics of remote sensing images in order to find out the important reasons for catastrophic forgetting. Specifically, we design a new model, named the TA model, which adds an auxiliary task. To ease the catastrophic forgetting problem, we propose diversity distillation loss to retain the knowledge of old tasks. Furthermore, we propose a general method for testing multihead classifiers using single-head evaluation. Extensive experiments on ISPRS Vaihingen and Potsdam benchmark demonstrate the effectiveness of our method. Although continuous learning realizes learning for continuously obtained labeled segmented data, it is difficult to obtain a large number of data labels. Few-shot learning uses prior knowledge and can quickly generalize to new tasks that contain only a small number of samples with supervision information. As for future work, we prepare to research few-shot continuous learning based on metric learning to achieve incremental learning with a small amount of new data.

References

[1] L. van der Maaten, "Accelerating t-SNE using tree-based algorithms," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 3221–3245, Oct. 2014.

[2] K. James *et al.*, "Overcoming catastrophic forgetting in neural networks," *Proc. Nat. Acad. Sci. USA*, vol. 114, no. 13, pp. 3521–3526, Mar. 2017.

[3] M. McCloskey and N. J. Cohen, "Catastrophic interference in connectionist networks: The sequential learning problem," *Psychol. Learn. Motivat.*, vol. 24, pp. 109–165, Dec. 1989.

[4] I. J. Goodfellow, M. Mirza, D. Xiao, A. Courville, and Y. Bengio, "An empirical investigation of catastrophic forgetting in gradient-based neural networks," 2013, *arXiv:1312.6211*.

[5] M. Masana, X. Liu, B. Twardowski, M. Menta, A. D. Bagdanov, and J. van de Weijer, "Class-incremental learning: Survey and performance evaluation on image classification," 2020, *arXiv:2010.15277*.

[6] N. Ammour, Y. Bazi, H. Alhichri, and N. Alajlan, "Continual learning approach for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, early access, Sep. 2, 2020, doi: 10.1109/LGRS.2020.3019071.

[7] O. Tasar, Y. Tarabalka, and P. Alliez, "Incremental learning for semantic segmentation of large-scale remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3524–3537, Sep. 2019.

[8] D. Isele and A. Cosgun, "Selective experience replay for lifelong learning," in *Proc. AAAI Conf. Artif. Intell.*, 2018, vol. 32, no. 1.

[9] Y. Feng, X. Sun, W. Diao, J. Li, X. Gao, and K. Fu, "Continual learning with structured inheritance for semantic segmentation in aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, early access, May 10, 2021, doi: 10.1109/TGRS.2021.3076664.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

[10] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 12, pp. 2935–2947, Dec. 2018.

[11] D. Lopez-Paz and M. Ranzato, "Gradient episodic memory for continual learning," 2017, *arXiv:1706.08840*.

[12] A. Chaudhry, P. K. Dokania, T. Ajanthan, and P. H. Torr, "Riemannian walk for incremental learning: Understanding forgetting and intransigence," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 532–547.

[13] M. Riemer *et al.*, "Learning to learn without forgetting by maximizing transfer and minimizing interference," 2018, *arXiv:1810.11910*.

[14] C. V. Nguyen, Y. Li, T. D. Bui, and R. E. Turner, "Variational continual learning," 2017, *arXiv:1710.10628*.

[15] G. M. Van de Ven and A. S. Tolias, "Three scenarios for continual learning," 2019, *arXiv:1904.07734*.

[16] M. De Lange *et al.*, "A continual learning survey: Defying forgetting in classification tasks," 2019, *arXiv:1909.08383*.

[17] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "ICaRL: Incremental classifier and representation learning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2001–2010.

[18] Y. Wu *et al.*, "Large scale incremental learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 374–382.

[19] U. Michieli and P. Zanuttigh, "Incremental learning techniques for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 3205–3212.

[20] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.

[21] Y. Feng, X. Sun, W. Diao, J. Li, and X. Gao, "Double similarity distillation for semantic image segmentation," *IEEE Trans. Image Process.*, vol. 30, pp. 5363–5376, 2021.

[22] O. Russakovsky *et al.*, "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, Dec. 2015.

[23] E. Belouadah, A. Popescu, and I. Kanellos, "Initial classifier weights replay for memoryless class incremental learning," 2020, *arXiv:2008.13710*.

[24] R. Aljundi, M. Rohrbach, and T. Tuytelaars, "Selfless sequential learning," 2018, *arXiv:1806.05421*.

[25] G. I. Parisi, R. Kemker, J. L. Part, C. Kanan, and S. Wermter, "Continual lifelong learning with neural networks: A review," *Neural Netw.*, vol. 113, pp. 54–71, May 2019.

[26] X. Liu, M. Masana, L. Herranz, J. Van de Weijer, A. M. Lopez, and A. D. Bagdanov, "Rotate your networks: Better weight consolidation and less catastrophic forgetting," in *Proc. 24th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2018, pp. 2262–2268.

[27] S.-W. Lee, J.-H. Kim, J. Jun, J.-W. Ha, and B.-T. Zhang, "Overcoming catastrophic forgetting by incremental moment matching," 2017, *arXiv:1703.08475*.

[28] A. Chaudhry, M. Ranzato, M. Rohrbach, and M. Elhoseiny, "Efficient lifelong learning with A-GEM," 2018, *arXiv:1812.00420*.

[29] A. Mallya and S. Lazebnik, "PackNet: Adding multiple tasks to a single network by iterative pruning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7765–7773.

[30] F. M. Castro, M. J. Marín-Jiménez, N. Guil, C. Schmid, and K. Alahari, "End-to-end incremental learning," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 233–248.

[31] P. Dhar, R. V. Singh, K.-C. Peng, Z. Wu, and R. Chellappa, "Learning without memorizing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5138–5146.

[32] K. Shmelkov, C. Schmid, and K. Alahari, "Incremental learning of object detectors without catastrophic forgetting," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 3400–3409.

[33] X. Liu, H. Yang, A. Ravichandran, R. Bhotika, and S. Soatto, "Multi-task incremental learning for object detection," 2020, *arXiv:2002.05347*.

[34] F. Cermelli, M. Mancini, S. Rota Bulo, E. Ricci, and B. Caputo, "Modeling the background for incremental learning in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9233–9242.

[35] G. Cheng, Z. Li, J. Han, X. Yao, and L. Guo, "Exploring hierarchical convolutional features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6712–6722, Nov. 2018.

[36] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.

[37] Y. Gao *et al.*, "Hyperspectral and multispectral classification for coastal wetland using depthwise feature interaction network," *IEEE Trans. Geosci. Remote Sens.*, early access, Jul. 26, 2021, doi: 10.1109/TGRS.2021.3097093.

[38] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, early access, Jul. 13, 2021, doi: 10.1109/TGRS.2021.3093334.

[39] H. Ding, X. Jiang, B. Shuai, A. Q. Liu, and G. Wang, "Semantic correlation promoted shape-variant context for segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8885–8894.

[40] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[41] M. Fan *et al.*, "Rethinking BiSeNet for real-time semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 9716–9725.

[42] Z. Liu *et al.*, "Swin transformer: Hierarchical vision transformer using shifted Windows," 2021, *arXiv:2103.14030*.

[43] S. Zheng *et al.*, "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 6881–6890.

[44] G. Deng, Z. Wu, C. Wang, M. Xu, and Y. Zhong, "CCANet: Class-constraint coarse-to-fine attentional deep network for subdecimeter aerial image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, early access, Feb. 12, 2021, doi: 10.1109/TGRS.2021.3055950.

[45] M. Zhang, W. Li, Q. Du, L. Gao, and B. Zhang, "Feature extraction for classification of hyperspectral and LiDAR data using patch-to-patch CNN," *IEEE Trans. Cybern.*, vol. 50, no. 1, pp. 100–111, Jan. 2020.

[46] H. Ding, X. Jiang, B. Shuai, A. Q. Liu, and G. Wang, "Context contrasted feature and gated multi-scale aggregation for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2393–2402.

[47] R. Niu, X. Sun, Y. Tian, W. Diao, K. Chen, and K. Fu, "Hybrid multiple attention network for semantic segmentation in aerial images," *IEEE Trans. Geosci. Remote Sens.*, early access, Mar. 24, 2021, doi: 10.1109/TGRS.2021.3065112.

[48] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[49] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.

[50] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[51] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Germany: Springer-Verlag, 2015, pp. 234–241.

[52] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*. Germany: Springer-Verlag, 2018, pp. 3–11.

[53] K. Yue, L. Yang, R. Li, W. Hu, F. Zhang, and W. Li, "TreeUNet: Adaptive tree convolutional neural networks for subdecimeter aerial image segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 156, pp. 1–13, Oct. 2019.

[54] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[55] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[56] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2017.

[57] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[58] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[59] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "High-resolution aerial image labeling with convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7092–7103, Dec. 2017.

[60] M. Volpi and D. Tuia, "Dense semantic labeling of subdecimeter resolution images with convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 881–893, Feb. 2016.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.

[61] J. Sherrah, "Fully convolutional networks for dense semantic labelling of high-resolution aerial imagery," 2016, *arXiv:1606.02585*.
[62] D. Marcos, M. Volpi, B. Kellenberger, and D. Tuia, "Land cover mapping at very high resolution with rotation equivariant CNNs: Towards small yet accurate models," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 96–107, Nov. 2018.
[63] G. Csurka, D. Larlus, and F. Perronnin, "What is a good evaluation measure for semantic segmentation?" in *Proc. Brit. Mach. Vis. Conf.*, vol. 27, 2013, p. 5244.
[64] N. Audebert, B. Le Saux, and S. Lefèvre, "Semantic segmentation of earth observation data using multimodal and multi-scale deep networks," in *Proc. Asian Conf. Comput. Vis.* Germany: Springer-Verlag, 2016, pp. 180–196.
[65] S. R. Bulo, L. Porzi, and P. Kontschieder, "In-place activated BatchNorm for memory-optimized training of DNNs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5639–5647.
[66] A. Paszke *et al.*, "Automatic differentiation in pytorch," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS-Workshop)*, 2017.



Junxi Li (Graduate Student Member, IEEE) received the B.Sc. degree from Jilin University, Changchun, China, in 2019. She is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

Her research interests include computer vision, continual learning, and remote sensing image processing, especially in semantic segmentation.



Xian Sun (Senior Member, IEEE) received the B.Sc. degree from the Beijing University of Aeronautics and Astronautics, Beijing, China, in 2004, and the M.Sc. and Ph.D. degrees (five year Ph.D. Program) from the Institute of Electronics, Chinese Academy of Sciences (CAS), Beijing, in 2009.

He was a Visiting Scholar with Karlsruher Institut Technologie, Karlsruhe, Germany, in 2013. He is currently a Professor with the Aerospace Information Research Institute, CAS. His research interests include computer vision, geospatial data mining, and remote sensing image understanding.

Dr. Sun was a recipient of the Outstanding Science and Technology Achievement Prize of CAS in 2016 and the First Prize for the State Scientific and Technological Progress of China in 2019. He also serves as an Associate Editor for IEEE ACCESS and a Guest Editor for the Special Issue of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS) and other journals.



Wenhui Diao received the B.Sc. degree from Xidian University, Xi'an, China, in 2011, and the M.Sc. and Ph.D. degrees (five year Ph.D. Program) from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2016.

He is currently an Associate Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision and remote sensing image analysis.



Peijin Wang received the B.Sc. degree from Tianjin University, Tianjin, China, in 2017, and the M.Sc. degree from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2020.

She is currently an Assistant Researcher with the Aerospace Information Research Institute, Chinese Academy of Sciences. Her research interests include computer vision, deep learning, and remote sensing.



Yingchao Feng (Graduate Student Member, IEEE) received the B.Sc. degree from Xidian University, Xi'an, China, in 2017. He is currently pursuing the Ph.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

His research interests include computer vision and deep learning, especially in object detection, semantic segmentation, and remote sensing.



Xiaonan Lu (Graduate Student Member, IEEE) received the B.Sc. degree from Xidian University, Xi'an, China, in 2019. He is currently pursuing the Ph.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, and the University of Chinese Academy of Sciences, Beijing.

His research interests include computer vision, pattern recognition, and remote sensing image processing, especially in image scene classification and object detection.



Guangluan Xu received the B.Sc. degree from Beijing Information Science and Technology University, Beijing, China, in 2000, and the M.Sc. and Ph.D. degrees (five year Ph.D. Program) from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2005.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision and remote sensing image understanding.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:55:23 UTC from IEEE Xplore. Restrictions apply.